The 2023-09-08 Al Haouz, Morocco earthquake (M_w 6.8, us7000kufc): a reproducible MTUQ + WISP validation study of a shallow continental thrust

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MTUQ (point-source moment tensor) + WISP (kinematic finite fault); every parameter documented

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Abstract

We determine the point-source moment tensor (MTUQ) and a kinematic finite-fault model (WISP) of the 2023-09-08 22:11:01 UTC M_w 6.8 Al Haouz earthquake — a shallow (∼26 km) oblique-reverse thrust in the Moroccan High Atlas that killed ∼2,900 people. Unlike the mostly-unstudied events we analysed previously, this one has a USGS moment tensor and finite-fault model, so we use it to validate the pipeline and to document every data, frequency-band and processing parameter for reproducibility. Our double-couple moment tensor, 256/67/65 (M_w 6.95), lies 5.1° (Kagan angle) from the USGS W-phase solution; our WISP finite fault on the steep plane (peak slip 1.62 m, centroid ∼25 km, ∼39×38 km; with the `shift_match` cross-correlation step applied) closely reproduces the USGS model (1.88 m, 25 km, 45×30 km) on the steep plane (though the teleseismic fit marginally, non-decisively, favours the shallow conjugate), and gives an ordinary stress drop of ∼3 MPa. The study also yields a methodological result that mirrors our deep-event work: for a shallow source the joint body-plus-surface-wave inversion returns a confidently wrong (flipped, normal-faulting) mechanism, because surface waves are dip-slip-indeterminate at shallow depth (Kanamori & Given, 1981) and mismatched by 1-D dispersion; the body waves, through their depth phases, carry the correct dip-slip sense. Inverting body-only recovers the USGS mechanism.

Plain-language summary

A magnitude-6.8 earthquake struck the High Atlas mountains south of Marrakesh, on a reverse (compressional) fault about 26 km deep. Because this event is already well studied, we use it to check our analysis tools against the U.S. Geological Survey's published answers — and they agree closely. Along the way we found a trap specific to shallow earthquakes: the long-period "surface waves" cannot tell which way such a fault slipped and gave the opposite (pull-apart) answer; the shorter-period "body waves" gave the correct (push-together) answer. This is the mirror image of what we learned for very deep earthquakes, where the body waves were the unreliable ones.

1 Event and USGS benchmarks

The earthquake (31.058°N, 8.385°W) is an intra-continental oblique-reverse rupture in the High Atlas. The USGS W-phase moment tensor is 122/29/132 (shallow SE-dipping) with conjugate 255/69/69 (steep NW-dipping), 94% double couple, centroid 30.5 km; the Mwc (26 km) and Mwb (24 km) solutions and the 19-km catalogue hypocentre span a genuine depth debate. The USGS finite-fault model adopted the steep plane (strike 251°, dip 69°), 45×30 km, peak slip 1.88 m, depth 25 km. These are the numbers we reproduce.

2 Data and parameters (documented for reproducibility)

Determinism. FDSN data fetching is deterministic in its parameters but not across time (data-centre availability changes; transient request failures drop stations). The reproducible unit is therefore the archived waveforms plus the parameters below: re-running from the saved SAC files is byte-deterministic, while re-fetching yields a near-identical set. The inversions are deterministic (MTUQ is an exhaustive grid search; WISP's annealing reproduces to four decimals). Compute is CPU-only (no GPU): MTUQ parallelises with MPI (`mpirun -n 20`); WISP uses OpenMP (Green's functions and annealing) plus a `multiprocessing` pool for data processing.

item	value
WISP data	`ffm manage acquire -t body` (II, G, IU, GE), ∼150 broadband stations
MTUQ data	20 GSN broadband, 31–86°, reused from the WISP fetch; response via StationXML inventory (output DISP), rotated ZNE→RT
MTUQ body band	15–40 s (`--bwband 15,40,80,8`); 80-s window, ±8 s shifts
MTUQ surface band	50–100 s (shown to flip the shallow mechanism; excluded from the solution)
MTUQ Green's functions	syngine ak135 (`ak135f_2s`)
MTUQ grid	DoubleCoupleGridRegular, 40 pts/axis (∼64k orientations × 5 magnitudes)
WISP body filter	0.006–1.0 Hz, wavelet scales 2–8
WISP surface waves	used (shallow source; above the 125-km surface-wave GF cap)
adopted depth	26 km

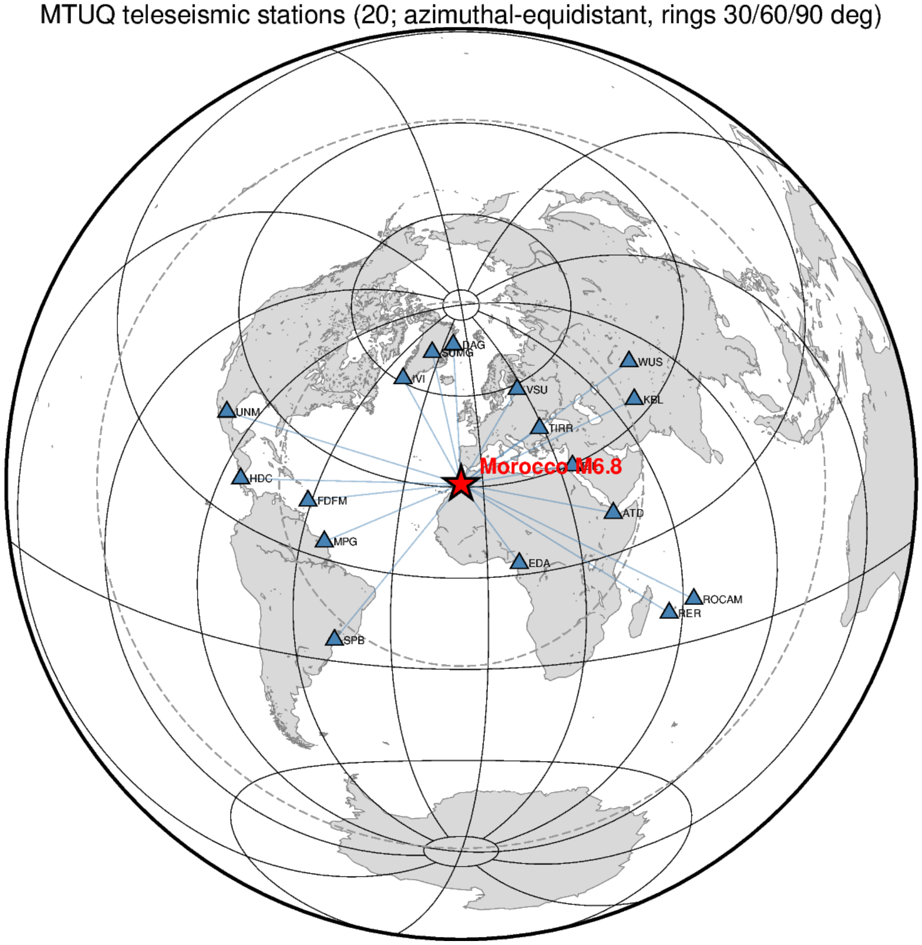

Figure 1: The 20 GSN teleseismic broadband stations used for the moment tensor, on an azimuthal-equidistant projection centred on the epicentre (rings at 30/60/90°).

3 Moment tensor: the shallow-event trap and cure

Run jointly (body 10–33 s + surface 50–100 s), the DC search converges confidently on a flipped, normal-faulting mechanism ~89° (Kagan) from USGS. The cause is physical: at shallow depth the vertical dip-slip moment-tensor elements (M_{rt}, M_{rp}) are weakly excited at long period — the Kanamori–Given shallow indeterminacy — and the surface waves are additionally mismatched by 1-D dispersion (large time shifts, poor amplitude fit), so they cannot fix, and actively mislead, the dip-slip sense. The body waves, through their depth phases, resolve it. We therefore invert body-only (a `--onlybody` option added to the runners). This is the mirror image of the deep-event rule, where short-period body waves fail and the surface/long-period waves are reliable.

The Kanamori–Given shallow indeterminacy — what it is, and how common

A moment tensor has six independent numbers. Two of them, M_{rt} and M_{rp} (the "vertical dip-slip" components), describe shear on a near-horizontal plane with a vertical slip direction. The Earth's surface is stress-free, and the long-period surface-wave radiation from these two components vanishes as the source approaches the free surface. So for a shallow earthquake, long-period surface waves carry almost no information about M_{rt} and M_{rp} — yet those are exactly the components that set the dip and the up-versus-down sense of a dip-slip fault. Long-period surface waves therefore cannot distinguish a shallow thrust from its flipped, normal-faulting counterpart: the strike and the scalar moment (the M_{rr}-dominated part) stay well-resolved, but the dip-slip sense lies in a near-null direction of the inversion. This is the Kanamori & Given (1981) result.

Why it flips rather than merely being uncertain: with the dip-slip sense nearly unconstrained, any small systematic data error tips the balance. Here the culprit is the 1-D velocity model, which mis-predicts surface-wave dispersion by a few seconds; the search then prefers whichever mechanism absorbs that mismatch with marginally lower L_2 misfit — and it picked the flipped one confidently (a deep, narrow minimum). That false confidence is the dangerous part: the fit looks good and the wrong answer looks certain.

Does it happen often? Yes — routinely, for most crustal earthquakes. Any shallow ($\lesssim$ few tens of km) dip-slip source, thrust or normal, is exposed to it whenever long-period surface waves dominate. It is why the Global CMT catalogue, for very shallow events, constrains the poorly-resolved vertical-dip-slip components (fixing $M_{rt} = M_{rp} = 0$ or clamping depth) and reports large dip uncertainties: for shallow events the robust outputs are strike and moment, not dip. How it is handled, in rough order: (1) use body waves, especially the depth phases pP and sP — their timing fixes depth and their amplitude/polarity fix the dip-slip sense surface waves cannot (the primary cure, and what `--onlybody` does); (2) constrain the source — impose a deviatoric or pure double-couple mechanism and/or fix depth, so the null directions cannot run away; (3) add near-field data — regional waveforms and static geodesy (InSAR/GPS), which sample the source directly and break the degeneracy (the vanishing-at-the-surface argument is a long-period far-field statement); (4) diagnose and report it — inspect the flat (null) direction of the misfit and treat a "confident" shallow surface-wave mechanism with suspicion. The reassuring part: this is a known, diagnosable trap, not a mysterious failure — once recognized, the body-only inversion gives the right answer (5.1° from USGS).

solution	M_w	strike/dip/rake	Kagan vs USGS Mww	%DC
DC (adopted)	6.95	256/67/65	5.1°	—
full moment tensor	6.80	270/63/63	19.1°	66% (USGS 94%)

Table 1: Body-only MTUQ. The DC solution matches USGS to 5.1°; the full MT is destabilised by the same shallow-source indeterminacy (spurious non-DC, 66% vs 94%), which is exactly why the DC constraint is standard for shallow sources. We adopt the DC solution.

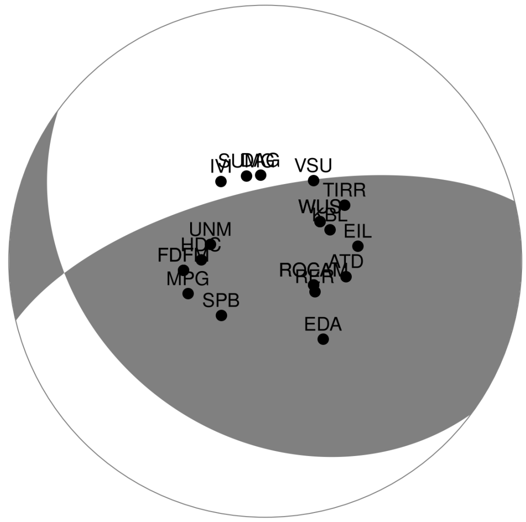

Figure 2: Adopted DC moment tensor (256/67/65) with the teleseismic station distribution — the correct oblique-reverse thrust.

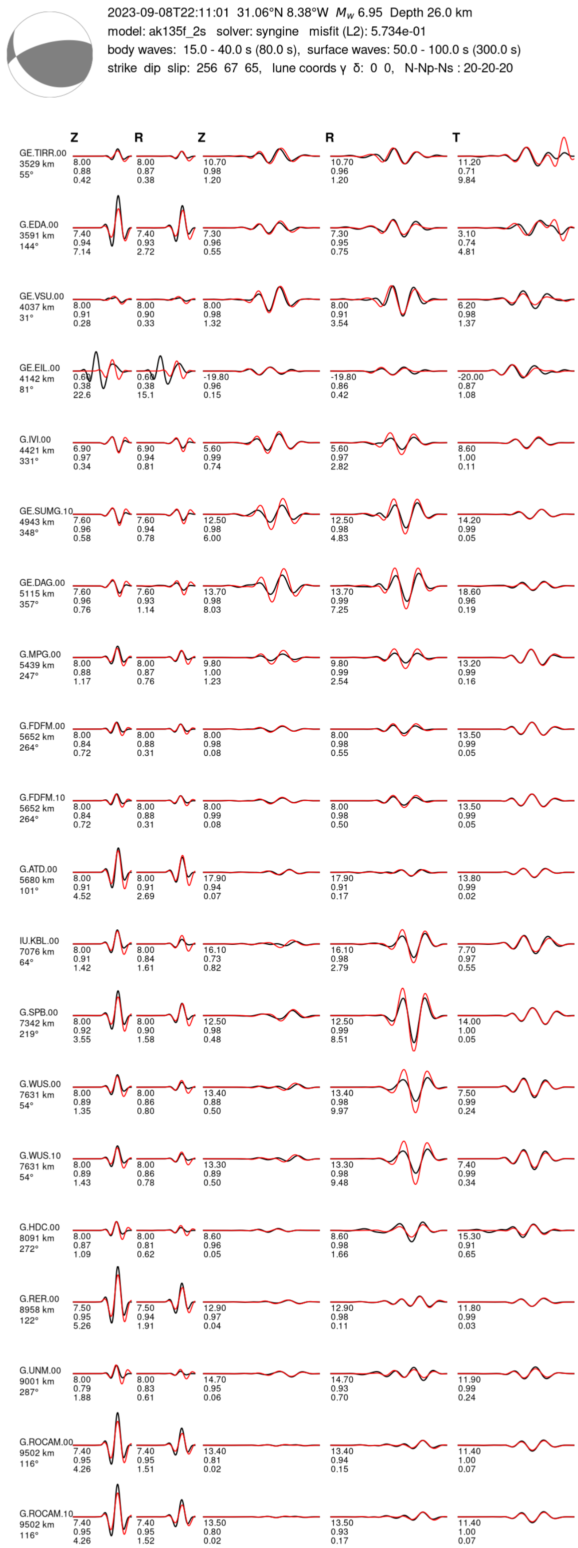

Figure 3: Fits of the adopted DC (body-only) solution (black observed, red synthetic). The body-wave windows were used in the inversion; the surface-wave windows are forward-predicted from the body-only mechanism — they were not inverted. That the body-derived mechanism, blind to the surface waves, nonetheless predicts them well confirms the surface waves were consistent with the correct solution all along; they simply could not, by themselves, choose the dip-slip sense (the Kanamori–Given indeterminacy above).

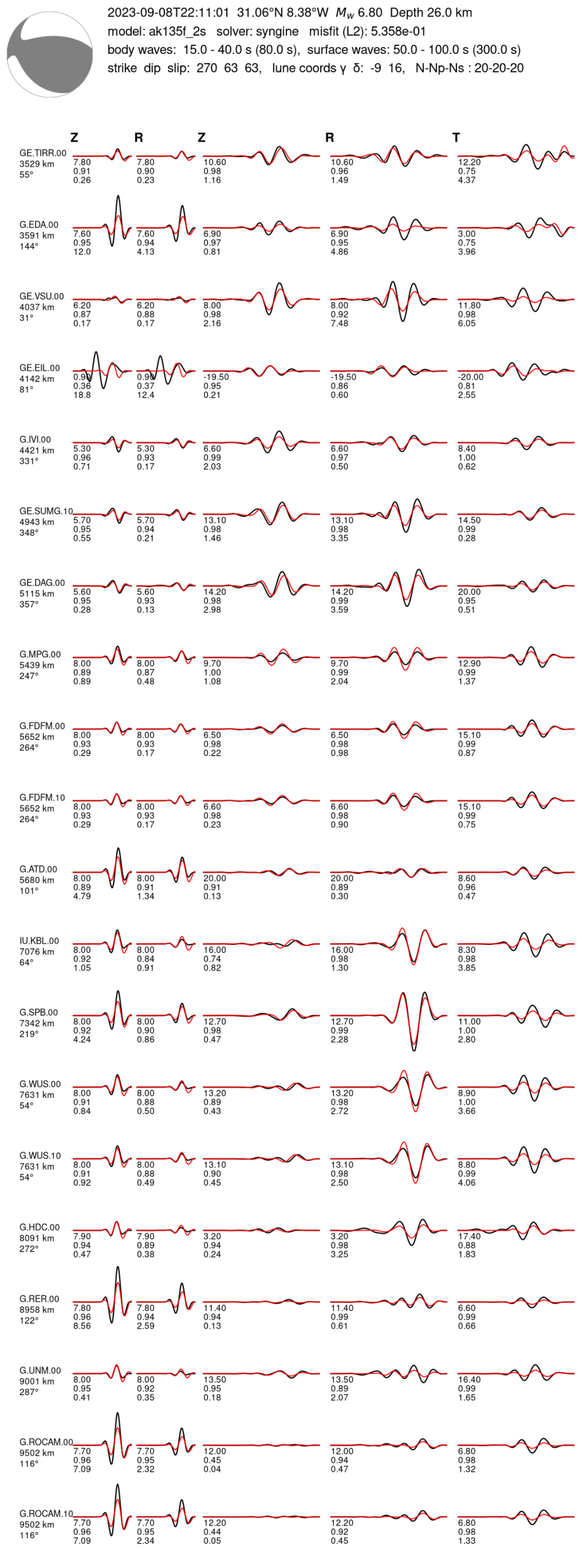

Figure 4: Body-wave fits of the full moment tensor (270/63/63, 66% DC), shown for completeness; the adopted solution is the better-constrained DC.

4 Depth

The catalogue (19 km), Mwc (26 km), USGS finite fault (25 km) and Mww centroid (30.5 km) disagree. A joint depth search is unreliable here (the surface-wave flip contaminates it), so we run a body-only DC+depth search: its global misfit minimum is at ∼33 km (a broad 30–34 km trough), independently supporting the deeper centroid (∼30–33 km, matching the USGS Mww 30.5 km) over the 19-km catalogue hypocentre, with a weaker secondary minimum near 10–14 km from the depth-phase trade-off. The WISP slip (below) spreads over 17–37 km, consistent with this mid-to-deep-crustal centroid.

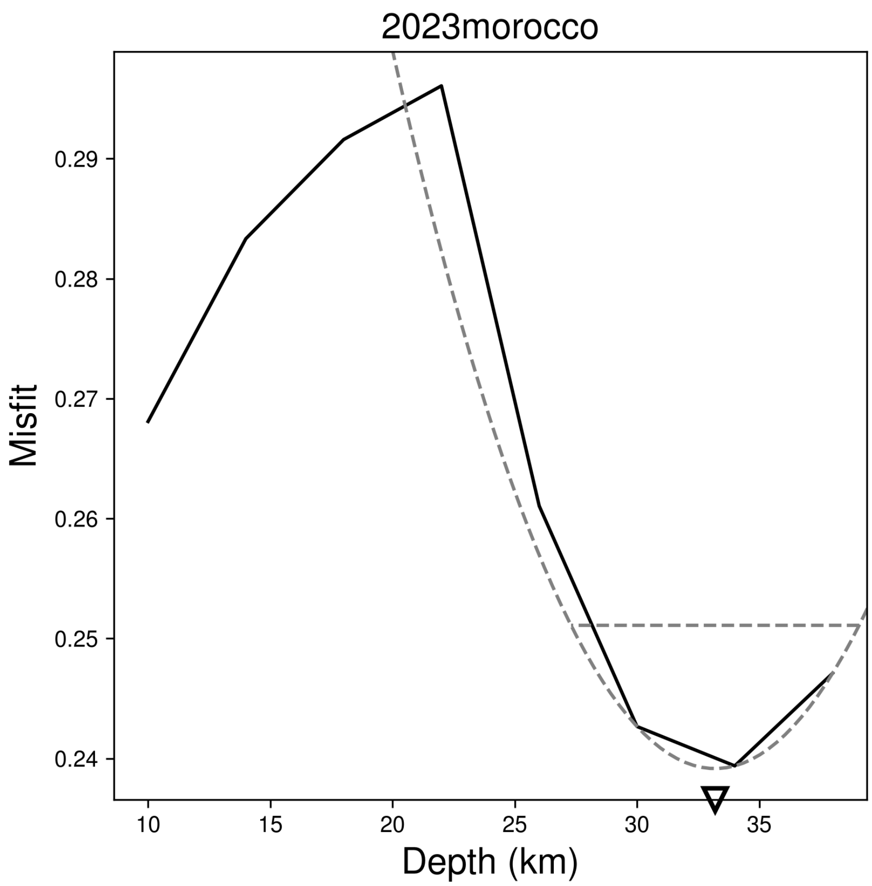

Figure 5: Body-only misfit versus source depth (best ∼33 km).

5 Finite fault: validation against the USGS model

We ran WISP `auto_model` on both nodal planes of our own MTUQ mechanism (internal consistency: the fault planes come from our moment tensor, not an external CMT). Being shallow, surface waves are fully used, together with teleseismic P (35 channels) and SH (26 channels) body waves. The adopted models apply the cross-correlation time-shift step (`shift_match`; see below), which every other event in this study used but the first Morocco pass had skipped.

plane	misfit (pre → post shift)	peak slip	$L \times W$	USGS finite fault
NP1 = steep 256/67 (= USGS FF plane)	0.1402 → 0.1235	1.50 → 1.62 m	39×38 km	251/69, 1.88 m, 25 kn
NP2 = shallow 126/33 (WISP-preferred)	0.1376 → 0.1196	1.46 → 1.71 m	43×38 km	—

Table 2: Plane discrimination is an honest ambiguity: after the shift WISP's teleseismic fit still marginally prefers the shallow plane (0.1196 vs 0.1235, ∼3%), disagreeing with USGS's finite-fault steep-plane choice — the shift does not flip the preference. As for the Chile events, teleseismic fit is a weak plane discriminant; USGS's steep choice rests on aftershocks and InSAR. We show the steep plane (NP1, = the USGS finite-fault plane) for validation.

The cross-correlation time-shift step

WISP provides `shift_match2` (auto cross-correlation, ±5 s body / ±12 s surface) and `manual_shift` (user-entered per-station) to remove the residual timing mismatch left by a 1-D velocity model and centroid mislocation. Each sets a trace's alignment index once and holds it fixed during the annealing (it is not a free inversion parameter). Applied here as auto cross-correlation for body (P+SH) and surface (Rayleigh+Love), then re-inverting, it cuts the misfit ∼12–13% on both planes and visibly aligns the SH peaks (TLY, YAK, KBS, PALK) while leaving M_w unchanged and merely sharpening the slip.

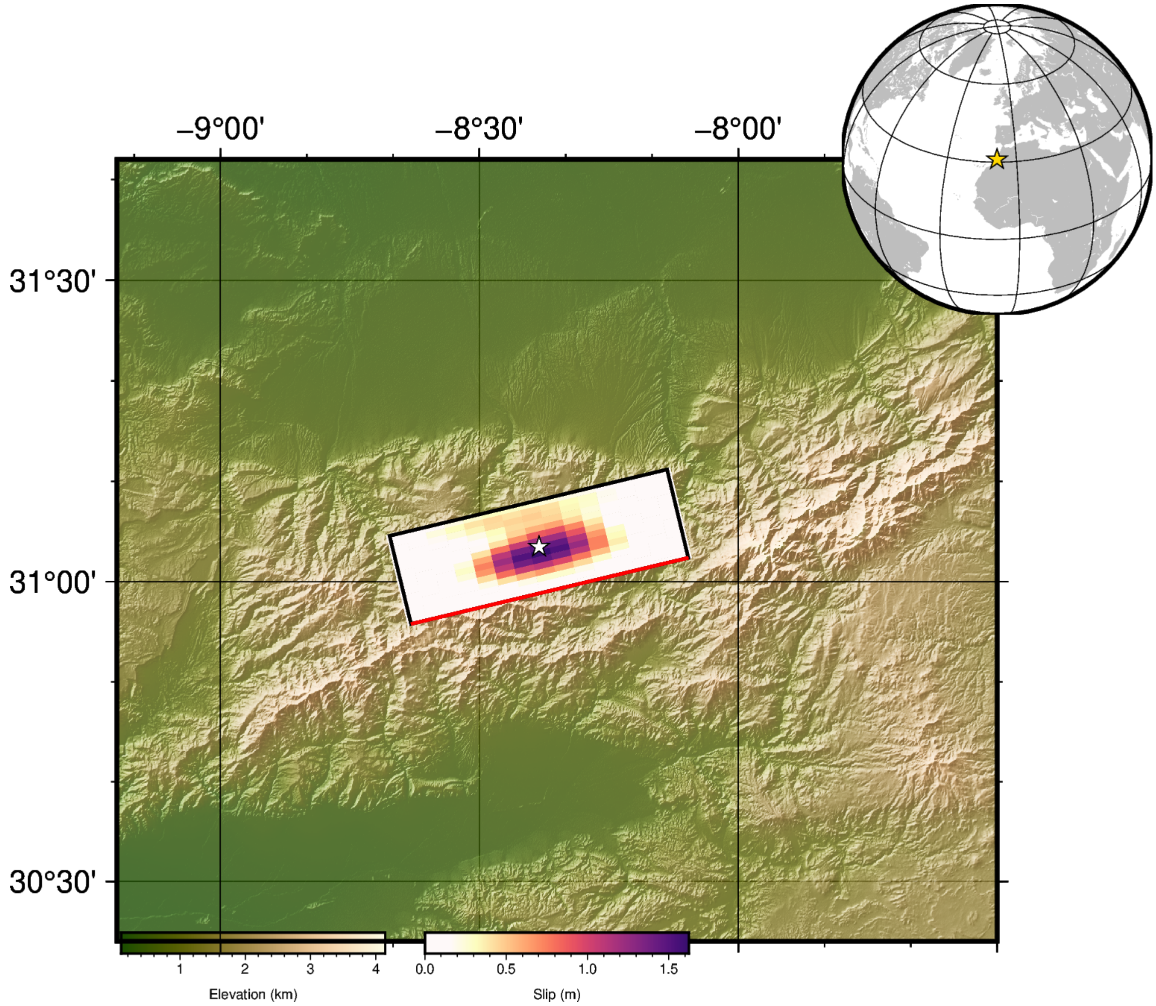

Figure 6: Map view of the preferred (steep-plane) slip model: fault slip projected to the surface, with the epicentre (star) and coastlines. WISP `PlotMap`.

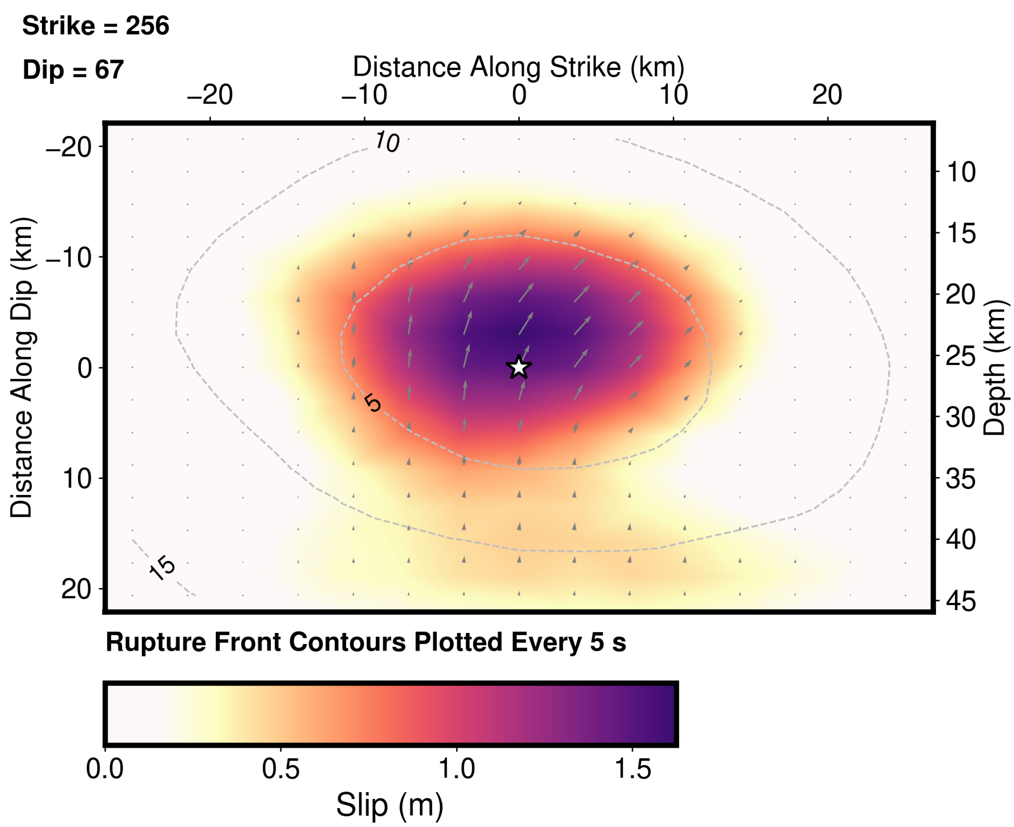

Figure 7: On-fault slip model (steep plane, with time shift). Star: hypocentre; contours: rupture front every 5 s; arrows: slip direction (reverse). Peak 1.62 m, centroid ∼25 km depth.

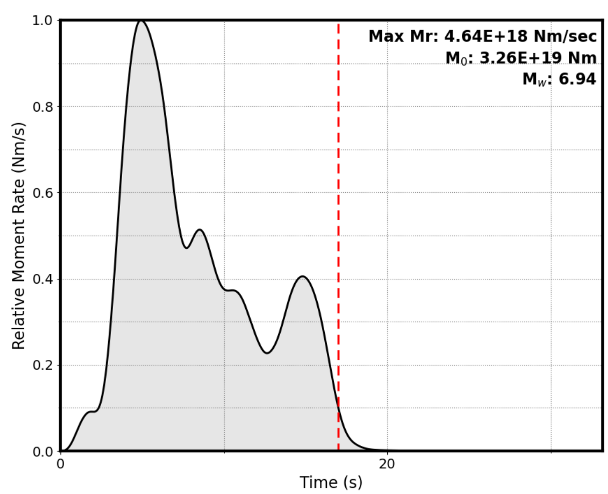

Figure 8: Moment-rate function (steep plane).

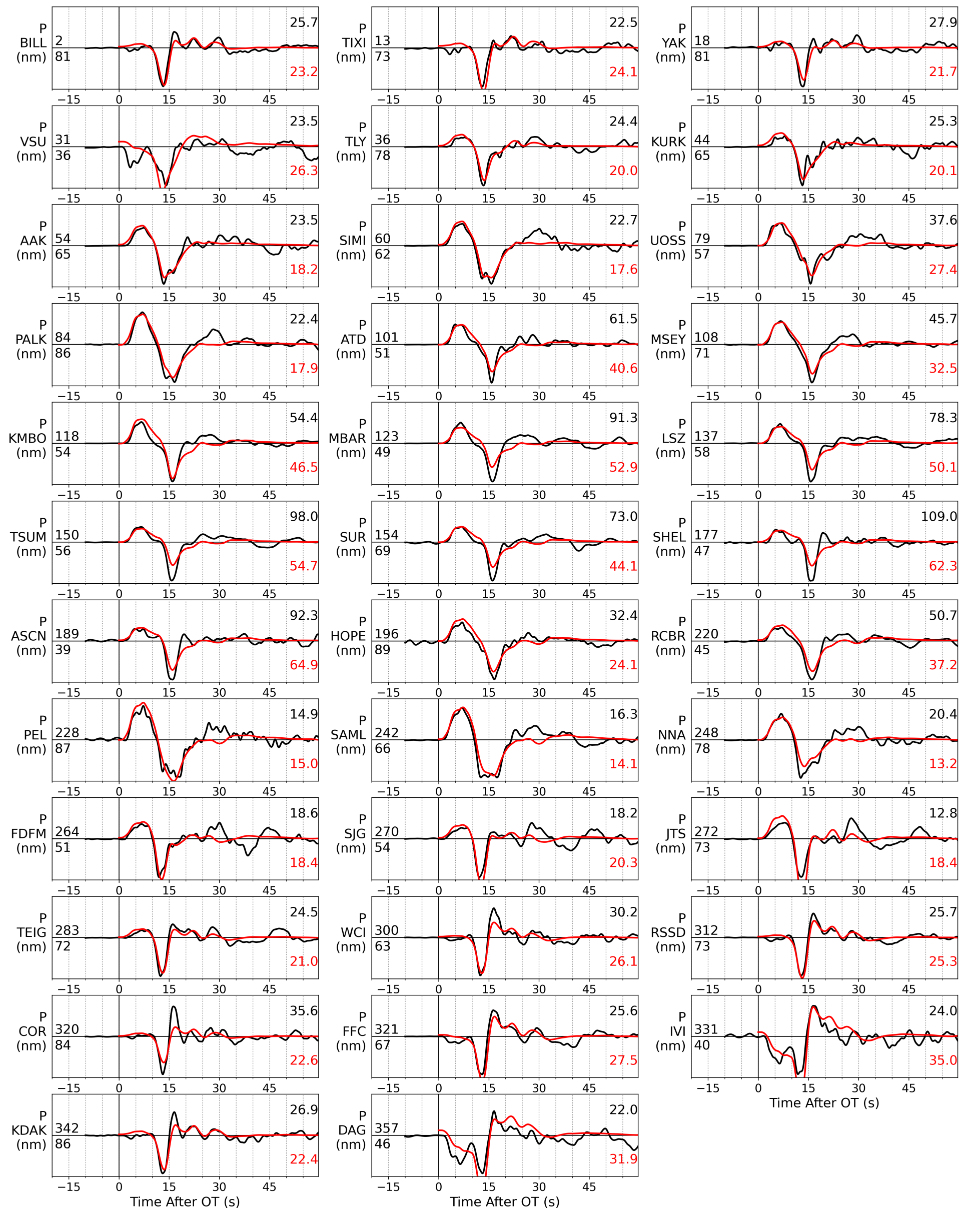

Figure 9: WISP P-wave fits (steep plane, with time shift).

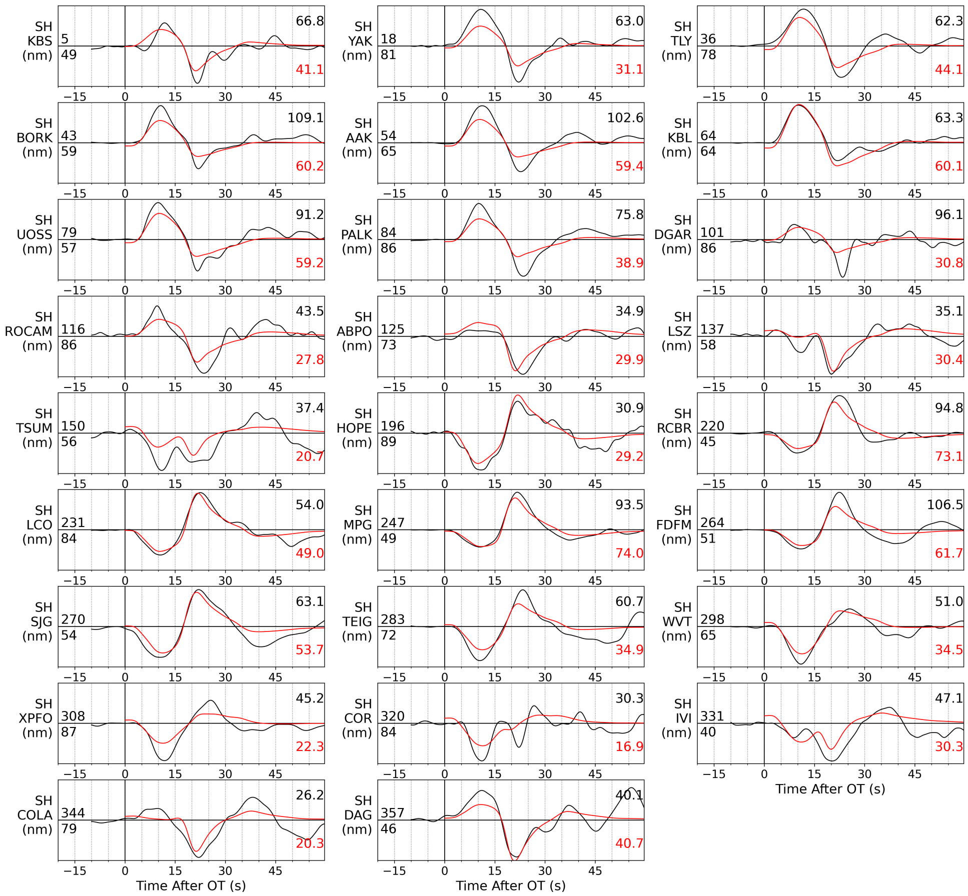

Figure 10: WISP SH-wave fits (steep plane, with time shift): teleseismic SH (transverse, 26 channels) alongside the 35 P channels. Compared to the un-shifted run, the synthetic peaks now align with the observed at TLY, YAK, KBS and PALK.

Figure 11: WISP Rayleigh-wave fits (steep plane).

Stress drop and moment magnitude

From the shifted slip distribution (Eshelby circular crack $\Delta\sigma = \frac{7}{16} M_0/R^3$, effective area from the slip participation ratio), the stress drop is 3.15 MPa (steep) / 3.26 MPa (shallow) — ∼3 MPa, an ordinary value for a continental crustal reverse earthquake (global median ∼3–4 MPa), independent of the plane. Both the MT and the finite fault give $M_w \approx 6.94$, i.e. ∼0.15 units (×1.6 in moment) above the USGS M_{ww} 6.8; the W-phase M_{ww} is the more authoritative long-period moment, and our body-wave-driven estimate runs higher. Since $\Delta\sigma \propto M_0$, normalising to M_{ww} would lower the stress drop to ∼2 MPa, so the defensible statement is ∼2–3 MPa, ordinary.

6 Rupture duration and a joint teleseismic + InSAR inversion

Our teleseismic model has a longer source ($T_{5-95} \approx 12$ s, tail to ∼17 s) than USGS (∼10–15 s); the moment-rate function shows a dominant pulse at ∼5 s plus a distinct second pulse at ∼15 s carrying our excess moment. From the USGS finite-fault product, their model uses a smaller fault (45×30 km vs our 54×44 km) and faster rupture velocity (2.64 vs 2.16 km/s); rise time and time-window count are essentially identical. Total STF ≈ (front crossing the fault) + rise, so our ×1.76 larger area and ×0.82 slower V_r multiply to ∼ ×1.6 in propagation time — exactly the ∼10 s →∼16 s difference. The long tail is weakly constrained: capping the source duration at 10 s re-inverts to an essentially identical misfit (0.1239 vs 0.1235).

Progressive evolution: adding geodetic data

USGS also inverted three Sentinel-1 interferograms. We fetched the same USGS resampled interferograms (1 ascending + 2 descending, 1,815 points; W. Barnhart) and re-inverted the steep plane jointly (teleseismic body + surface + InSAR), building WISP static Green's functions for the geodetic points (a per-track linear ramp is co-estimated). The seismic-only model (finite-fault section above) is kept unchanged for comparison.

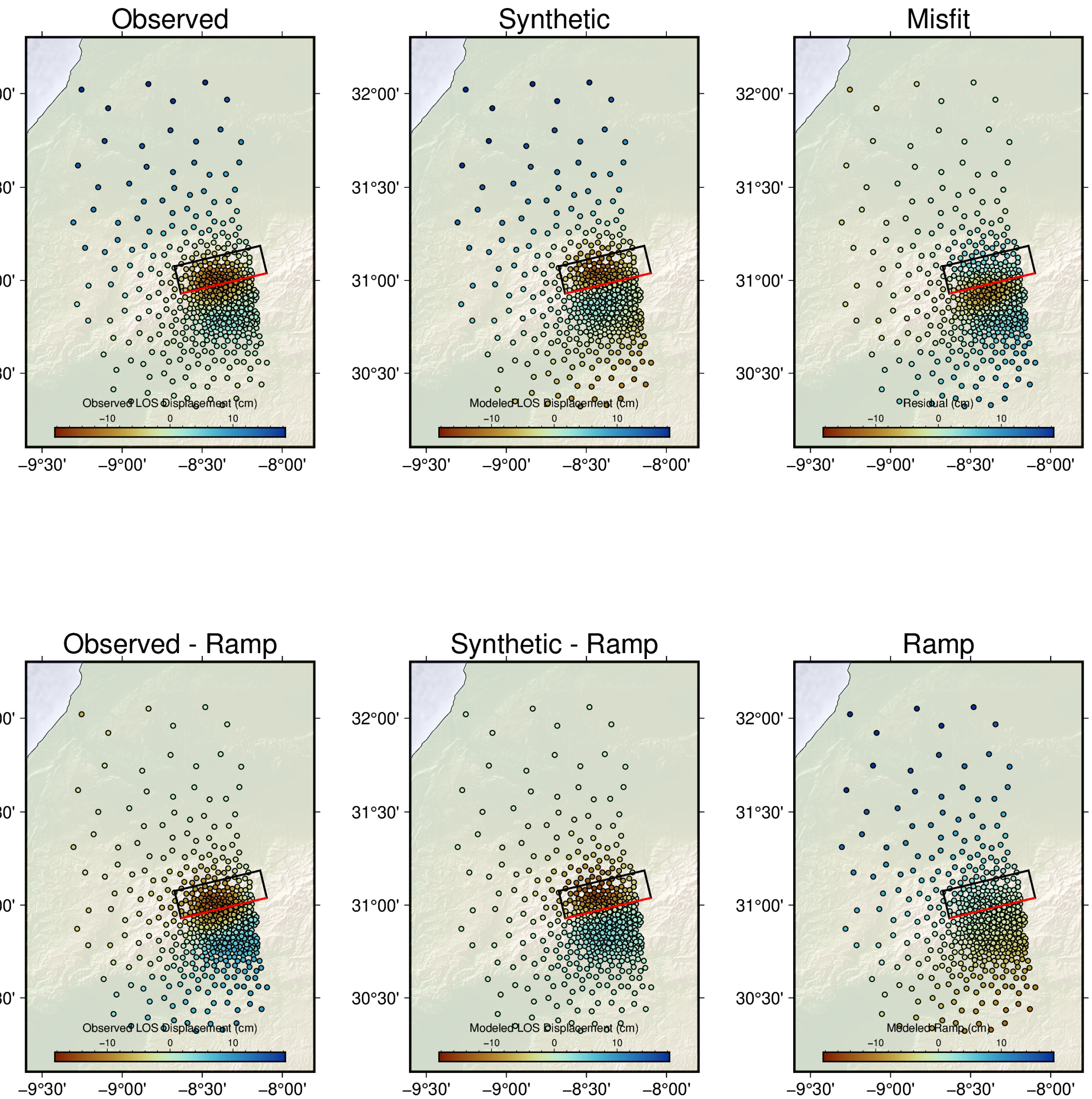

Figure 12: Joint-inversion InSAR fit, ascending track (observed / synthetic / residual; bottom row removes the co-estimated ramp). The near-fault co-seismic lobes are well reproduced.

	seismic-only	+ InSAR (joint)	USGS
teleseismic misfit	0.1235	0.1236 (unchanged)	—
InSAR penalization	(not used)	2.05 $\rightarrow$ 0.18 (now fit)	fit
M_w / M_0 (N m)	6.94 / 3.26e19	6.94 / 3.26e19 (unchanged)	6.84 / 2.27e19
duration T_{5-95}	12.3 s	12.3 s (unchanged)	$\sim$10–15 s
slip compactness N_{eff}	82/225	84/225 (unchanged)	—

Table 3: Adding the InSAR fits the interferograms (penalization 2.05→0.18) with no teleseismic penalty, but leaves moment, duration and extent unchanged — our seismic-only model already satisfies the geodesy. The duration and moment differences from USGS are therefore a parameterization effect (larger fault template + slower V_r + seed), not an InSAR-data effect: both models fit the same interferograms.

7 Summary

1. The pipeline reproduces the USGS moment tensor to 5.1° and the USGS finite-fault model (slip amplitude, depth, dimensions, preferred plane) for this well-studied event — a clean end-to-end validation across a new tectonic regime (shallow continental thrust).

2. Shallow-event rule: invert body waves for the mechanism; surface waves are dip-slip-indeterminate at shallow depth and flip the solution. Mirror of the deep-event rule.

3. Use the DC constraint for shallow sources — the full MT is destabilised by the same indeterminacy (66% vs USGS 94% DC).

4. Apply the **`shift_match`** cross-correlation step (as every other event did): it cut the WISP misfit $\sim$12–13%, aligned the SH peaks, and sharpened the slip without changing M_w. Stress drop $\sim$3 MPa (ordinary); M_w 6.94, $\sim$0.15 above the USGS M_{ww} 6.8.

5. Rupture duration vs USGS is a parameterization effect, not a data effect. A joint teleseismic+InSAR inversion using USGS's own interferograms fits the geodesy (penalization 2.05→0.18) but leaves our moment/duration/extent unchanged — the seismic model already satisfies the InSAR. The $\sim\times2$ duration and M_w 6.94 vs 6.84 come from the larger fault template + slower V_r + seed, not the InSAR.

6. Every parameter is tabulated; the analysis re-runs deterministically from the archived waveforms. Pipeline is CPU-parallel (MPI + OpenMP), no GPU.